Scattering part 1

Let $\psi(r,\theta,\phi)$ be a wave function of the form

$$\psi(r,\theta,\phi) = \frac{e^{ikr}}{r} f(\theta,\phi)$$

where

$$k = \frac{\sqrt{2mE}}{\hbar}$$

The density is

$$|\psi|^2 = \psi^*\psi = \frac{|f(\theta,\phi)|^2}{r^2}$$

Let $d\sigma$ be the differential cross section

$$d\sigma = |\psi|^2\,dA = |\psi|^2 r^2\,d\Omega = |f(\theta,\phi)|^2\,d\Omega$$

Hence

$$\frac{d\sigma}{d\Omega} = |f(\theta,\phi)|^2$$

For Coulomb potentials we have

$$f(\theta,\phi) = -\frac{Z\alpha\hbar c}{2E(1-\cos\theta)}$$

where Z is the dimensionless integer

$$Z = \frac{q_1 q_2}{e^2}$$

Hence the Rutherford cross section

$$\frac{d\sigma}{d\Omega} = \left[\frac{Z\alpha\hbar c}{2E(1-\cos\theta)}\right]^2$$

Noting that

$$(1-\cos\theta)^2 = 4\sin^4(\theta/2)$$

we can also write

$$\frac{d\sigma}{d\Omega} = \left(\frac{Z\alpha\hbar c}{2E}\right)^2 \frac{1}{4\sin^4(\theta/2)}$$

Verify dimensions of exponential.

$$\frac{\sqrt{2mE}}{\hbar} r = \frac{\sqrt{[\mathrm{kg}]\,[\mathrm{J}]}}{[\mathrm{J\,s}]}\,[\mathrm{m}] = [1] \tag{1}$$

Verify dimensions of cross section.

$$\left(\frac{Z\alpha\hbar c}{2E}\right)^2 = \left(\frac{[\mathrm{J\,s}]\,[\mathrm{m\,s^{-1}}]}{[\mathrm{J}]}\right)^2 = [\mathrm{m}^2] \tag{2}$$

Eigenmath script